

Bug Fix Log — Grading Console

All bugs were found in the original prototype and verified after fixing. Live app: <https://grading-console.pages.dev>

#	Bug / Issue Identified	How You Reproduced It	Root Cause	Fix Implemented	How You Tested the Fix
1	Course list duplicates after uploading another file	Uploaded two Excel files one after another; dropdown grew to 312 options and still listed old courses	Existing course options were never cleared before adding new ones	Rebuilt the course list from scratch on every import; kept the selected course only if it still exists	Uploaded 2 files and verified the list showed only the new file's unique courses
2	Average shows NaN when a course has no students	Selected a course left over from the previous file	Average was calculated on an empty dataset (dividing by 0)	Added empty-data handling: shows “—” and a “No gradeable records” message	Tested an empty course; no NaN shown
3	Every BITS ID exported as “undefined”	Uploaded a file with header “Student’s BITS ID” and exported	Code read column “BITS ID”, but the real header is “Student’s BITS ID”	Matched columns by normalised name (ignores case, spaces and apostrophe style)	Exported CSV; every row showed the real BITS ID
4	Course dropdown shows one entry per student	Uploaded 248 rows for 3 courses and got 248 options	Added each row's course without removing duplicates	Built a unique, sorted course list with student counts	Verified exactly 3 courses (120 / 50 / 78)
5	Min and Max values swapped	Min box showed 94 and Max showed 47	HTML labels pointed at each other's element IDs	Rebuilt the stats panel with correct labels (Lowest / Highest)	Compared with the true sorted min and max
6	Students silently dropped when ranges don't cover 0-100	Set A's max to 90; no error, and only 48 of 50 students exported	Validation never checked that ranges start at 0 and end at 100	Checked every mark 0-100 and named each gap or overlap; blocked export if any student has no grade	Set A max = 90 → error “No grade covers marks 91-100”; export disabled
7	Invalid marks silently disappear from export	Used a file with -5, 104, 79.5, blank and “AB”; only 8 of 14 rows exported	No validation of the marks	Listed invalid rows under “Records requiring attention” with row number and reason	Edge-case file showed every issue; export required acknowledgement

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8	Decimal marks (e.g. 79.5) get no grade	Uploaded a mark of 79.5	It falls between bands 70-79 and 80-100	Flagged decimals; added an optional one-click "round half-up" that can be undone	79.5 → 80 → graded A; the change is listed
9	Duplicate BITS IDs exported twice	Same ID appeared twice with marks 71 and 64	No duplicate check	Exact duplicates counted once; conflicting marks held back for review	Duplicate appeared once; conflicting ones were flagged
10	Average and median wrong for marks stored as text	Traced the average/median code with marks like "68" stored as text in Excel	JavaScript joined the strings instead of adding the numbers	Converted numeric text to numbers	"70", "72", 50 → mean 64 and median 70 correct
11	Console error when clicking Reset before choosing a course	Clicked Reset, confirmed twice, and got a TypeError	The grade inputs don't exist yet	Reset only appears in the Grade step; replaced the 2 pop-ups with Undo	Reset and Undo work; no console errors
12	Corrupt / empty / wrong-column file gives no message	Uploaded a damaged file and a wrong-header file; nothing happened	No error handling around file reading	Clear messages that name the missing and found columns; the previous data is kept	Tested corrupt, empty, header-only, PDF and wrong-column files
13	.xlsx files hidden in the file picker	File input had accept=".xls"	Wrong file extension	Changed to accept .xlsx and .xls, and added drag-and-drop	Uploaded .xlsx successfully
14	Uploading the same file twice does nothing	Re-selected the same corrected file	Input value was never reset, so no change event fired	Reset the input after each upload	Same file imported twice; both imports registered
15	CSV breaks on commas and allows formula injection	Instructor "Rao, K." and ID =HYPERLINK(...)	No CSV quoting or escaping	Proper quoting; formulas neutralised with a leading '	CSV opened correctly with the values shown as text
16	Histogram bars overflow the chart	500-student course → a 1,488 px bar on a 240 px canvas	Fixed scale (count × 12)	Auto-scaled chart axis	20,000-row file; chart stayed in bounds

#	Bug / Issue Identified	How You Reproduced It	Root Cause	Fix Implemented	How You Tested the Fix
17	Bell curve breaks when all marks are equal	Everyone scored 75	Standard deviation = 0, so division by zero	Removed the bell curve; added Mean and Median markers	"Every student scored 75" shown correctly
18	Export not flagged after ranges change	Exported, then changed a cutoff	Nothing recorded what was exported	Added a "Changed since your last export" warning	Changed range after export; warning appeared